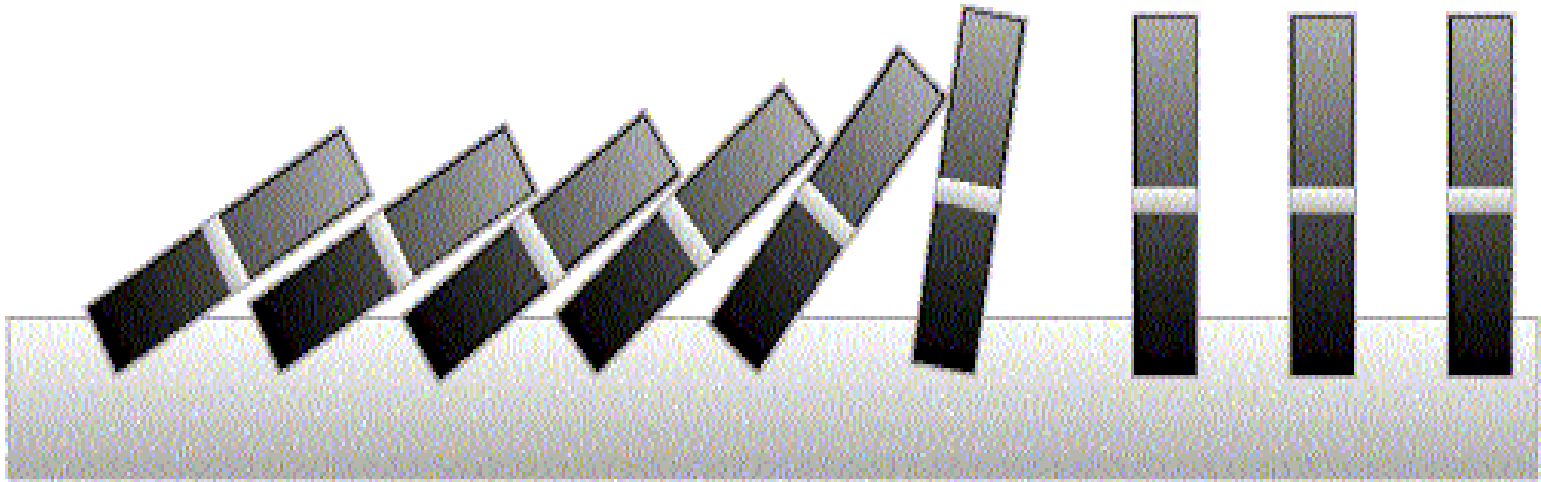


Mathematical Induction I



This Lecture

We have already discussed different proof techniques.

This time we will focus on probably the most important one

- mathematical induction.

This lecture's plan is to go through the following:

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

Why Induction

- We need to prove many statements that are true for a set (e.g., all integers):
 - For every positive integer n :
 - $n! \geq n^n$
 - $n^3 - n$ is divisible by 3
 - A set with n elements has 2^n subsets
 - The sum of first n positive integers is $n(n+1)/2$
 - etc.

Induction: Main Idea

- Proofs using mathematical induction have two parts (assuming the domain of positive integer).
 - First, they show that the statement holds for the positive integer 1.
 - Second, they show that if the statement holds for a positive integer then it must also hold for the next larger integer.
- Formally this is based on the rule of inference:
 - if $P(1)$ and $\forall k(P(k) \rightarrow P(k + 1))$ are true for the domain of positive integers, then $\forall n P(n)$ is true.

Main Idea of Induction

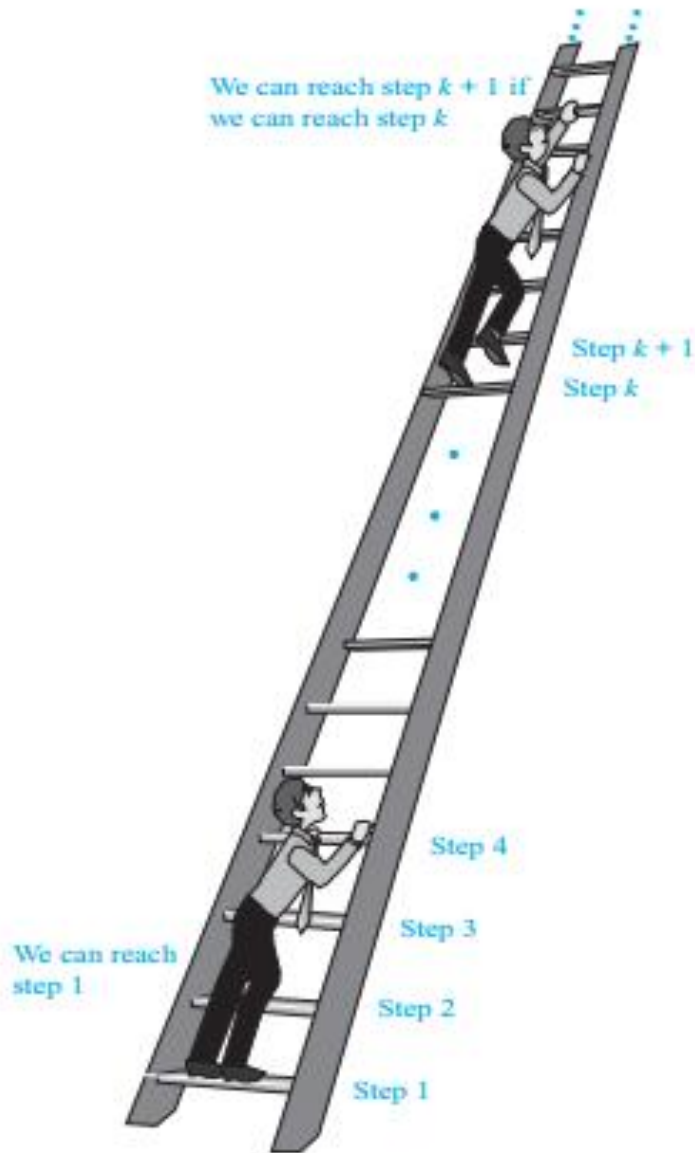


FIGURE 1 Climbing an Infinite Ladder.

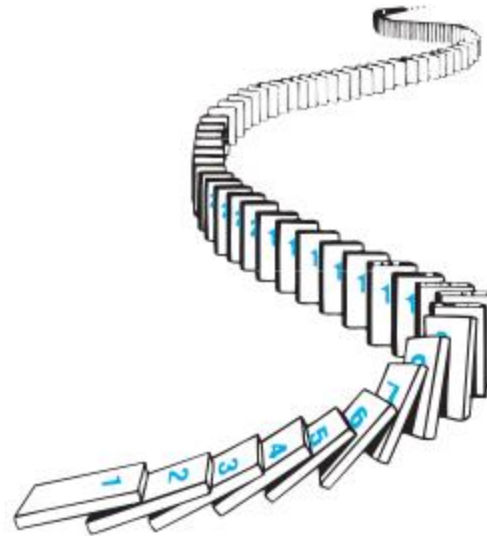


FIGURE 2 Illustrating How Mathematical Induction Works Using Dominoes.

Principle of MI

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

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If $P(k)$ is true then $P(k+1)$ is true

Principle of MI

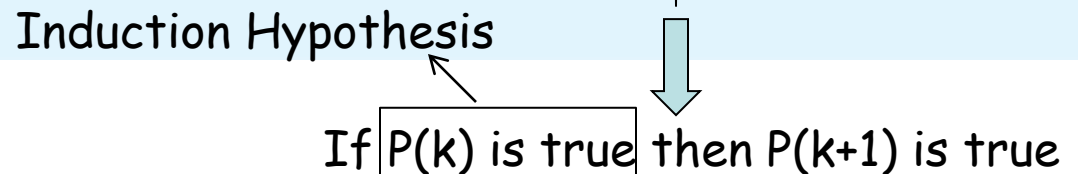
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Induction Hypothesis

If $P(k)$ is true then $P(k+1)$ is true



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Induction Hypothesis

If $P(k)$ is true then $P(k+1)$ is true

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k+1))) \rightarrow \forall n P(n)$$

Odd Powers Are Odd: Induction Idea

Fact: If m is odd and n is odd, then nm is odd.

Proposition: for an odd number m , m^i is odd for all non-negative integer i .

$$\forall i \in \mathbb{Z} \quad \text{odd}(m^i)$$

Let $P(i)$ be the proposition that m^i is odd.

$$\forall i \in \mathbb{Z} \quad P(i)$$

Idea of induction.

- $P(1)$ is true by definition.
- $P(2)$ is true by $P(1)$ and the fact.
- $P(3)$ is true by $P(2)$ and the fact.
- $P(i+1)$ is true by $P(i)$ and the fact.
- So $P(i)$ is true for all i .

Divisibility by a Prime: Induction Idea

Theorem. Any integer $n > 1$ is divisible by a prime number.

- Let n be an integer.
- If n is a prime number, then we are done.
- Otherwise, $n = ab$, both are smaller than n .
- If a or b is a prime number, then we are done.
- Otherwise, $a = cd$, both are smaller than a .
- If c or d is a prime number, then we are done.
- Otherwise, repeat this argument, since the numbers are getting smaller and smaller, this will eventually stop and we have found a prime factor of n .

Idea of induction.

Idea of Induction

Objective: Prove $\forall n \geq 0 \ P(n)$

This is to prove

$$\underline{P(0)} \wedge \underline{P(1)} \wedge \underline{P(2)} \wedge \dots \wedge \underline{P(n)} \dots$$

The diagram shows the sequence of propositions $P(0) \wedge P(1) \wedge P(2) \wedge \dots \wedge P(n) \dots$. Each term $P(k)$ is underlined. Red curved arrows connect the underlined terms, starting from $P(0)$ and pointing to $P(1)$, then from $P(1)$ to $P(2)$, and so on, illustrating the inductive step where the truth of $P(k)$ is used to prove $P(k+1)$.

The idea of induction is to first prove $P(0)$ unconditionally,
then use $P(0)$ to prove $P(1)$
then use $P(1)$ to prove $P(2)$
and repeat this to infinity...

The Induction Rule

0 and (from n to $n+1$),

proves 0, 1, 2, 3,

Very easy
to prove

Much easier to
prove with $P(n)$ as
an assumption.

valid rule

$$P(0), \forall n \in \underline{\mathbb{Z}} P(n) \Rightarrow P(n+1)$$

$$\forall m \in \underline{\mathbb{Z}}. P(m)$$

The point is to use the knowledge on smaller problems to solve bigger problems.



Good and Bad of MI

Good

- It can prove a conjecture once it has been made (and is true)
- Often simple

Bad

- Cannot find/propose a new theorem
- Do not provide any insight as to why the theorem is true

This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

Proving an Equality

$$r \neq 1 \quad \forall n \geq 1 \quad 1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

Let $P(n)$ be the induction hypothesis that the statement is true for n .

Base case: $P(1)$ is true because $\text{LHS} = 1 + r = \text{RHS} = (r^2 - 1)/(r - 1) = r + 1$

Induction step: assume $P(n)$ is true, prove $P(n+1)$ is true.

That is, assuming: $1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$

Want to prove: $1 + r + \dots + r^n + r^{n+1} = \frac{r^{n+2} - 1}{r - 1}$

This is much easier to prove than proving it directly,
because we already know the sum of the first n terms!

Proving an Equality

$$r \neq 1 \quad \forall n \geq 1 \quad 1 + r + \dots + r^n = \frac{r^{n+1} - 1}{r - 1}$$

Let $P(n)$ be the induction hypothesis that the statement is true for n .

Base case: $P(1)$ is true because $\text{LHS} = 1 + r = \text{RHS} = (r^2 - 1)/(r - 1) = r + 1$

Induction step: assume $P(n)$ is true, prove $P(n+1)$ is true.

$$\begin{aligned} & 1 + r + \dots + r^n + r^{n+1} \\ &= \frac{r^{n+1} - 1}{r - 1} + r^{n+1} \quad \text{by induction} \\ &= \frac{r^{n+1} - 1 + r^{n+2} - r^{n+1}}{r - 1} \\ &= \frac{r^{n+2} - 1}{r - 1} \end{aligned}$$

Proving an Equality

$$\forall n \geq 1 \quad 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$$

Let $P(n)$ be the induction hypothesis that the statement is true for n .

Base case: $P(1)$ is true

Induction step: assume $P(n)$ is true, prove $P(n+1)$ is true.

$$\begin{aligned} & 1^3 + 2^3 + \dots + n^3 + (n+1)^3 \\ &= \left(\frac{n(n+1)}{2}\right)^2 + (n+1)^3 \quad \text{by induction} \\ &= (n+1)^2(n^2/4 + n + 1) \\ &= (n+1)^2\left(\frac{n^2 + 4n + 4}{4}\right) = \left(\frac{(n+1)(n+2)}{2}\right)^2 \end{aligned}$$

Proving a Property

$$\forall n \geq 1, \quad 2^{2n} - 1 \text{ is divisible by } 3$$

Base Case ($n = 1$): $2^{2n} - 1 = 2^2 - 1 = 3$

Induction Step: Assume $P(i)$ for some $i \geq 1$ and prove $P(i + 1)$:

Assume $2^{2i} - 1$ is divisible by 3, prove $2^{2(i+1)} - 1$ is divisible by 3.

$$\begin{aligned} 2^{2(i+1)} - 1 &= 2^{2i+2} - 1 \\ &= 4 \cdot 2^{2i} - 1 \\ &= \underbrace{3 \cdot 2^{2i}} + \underbrace{2^{2i} - 1} \end{aligned}$$

Divisible by 3

Divisible by 3 by induction

Proving a Property

$$\forall n \geq 2, \quad n^3 - n \text{ is divisible by } 6$$

Base Case ($n = 2$): $2^3 - 2 = 6$

Induction Step: Assume $P(i)$ for some $i \geq 2$ and prove $P(i + 1)$:

Assume $n^3 - n$ is divisible by 6

Prove $(n + 1)^3 - (n + 1)$ is divisible by 6.

$$\begin{aligned}(n + 1)^3 - (n + 1) &= (n^3 + 3n^2 + 3n + 1) - (n + 1) \\ &= (n^3 - n) + 3(n^2 + n)\end{aligned}$$

Divisible by 6
by induction

Divisible by 2
by case analysis

HOW?

$n^2 + n$ is divisible by 2 (Case Analysis)

- Say n is odd:
 - Recall: for an odd number m , m^i is odd for all non-negative integer i .
 - So, n^2 is odd; n is also odd
 - $n^2 + n$ is sum of two odd numbers $\Rightarrow$ even
- Say n is even
 - Easy...

Proving an Inequality

$$\forall n \geq 3, \quad 2n + 1 < 2^n$$

Base Case ($n = 3$): $2n + 1 = 7 < 2^n = 2^3 = 8$

Induction Step: Assume $P(i)$ for some $i \geq 3$ and prove $P(i + 1)$:

Assume $2i + 1 < 2^i$, prove $2(i + 1) + 1 < 2^{(i+1)}$

$$2(i + 1) + 1 = \underline{2i + 1} + 2$$

$$< 2^i + 2 \quad \text{by induction}$$

$$< 2^i + 2^i \quad \text{since } i \geq 3, \text{ we must have } 2 < 2^i$$

$$= 2^{(i+1)}$$

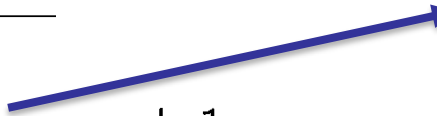
Proving an Inequality

$$\forall n \geq 2, \quad \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

Base Case ($n = 2$) is true [$1/\sqrt{1} + 1/\sqrt{2} = 1.71 > 1.41 = \sqrt{2}$]

Induction Step: Assume $P(i)$ for some $i \geq 2$ and prove $P(i + 1)$:

$$\begin{aligned} & \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} \\ & > \sqrt{n} + \frac{1}{\sqrt{n+1}} \quad \text{by induction} \\ & = \frac{\sqrt{n}\sqrt{n+1} + 1}{\sqrt{n+1}} \\ & > \frac{\sqrt{n}\sqrt{n} + 1}{\sqrt{n+1}} = \frac{n+1}{\sqrt{n+1}} \\ & = \sqrt{n+1} \end{aligned}$$

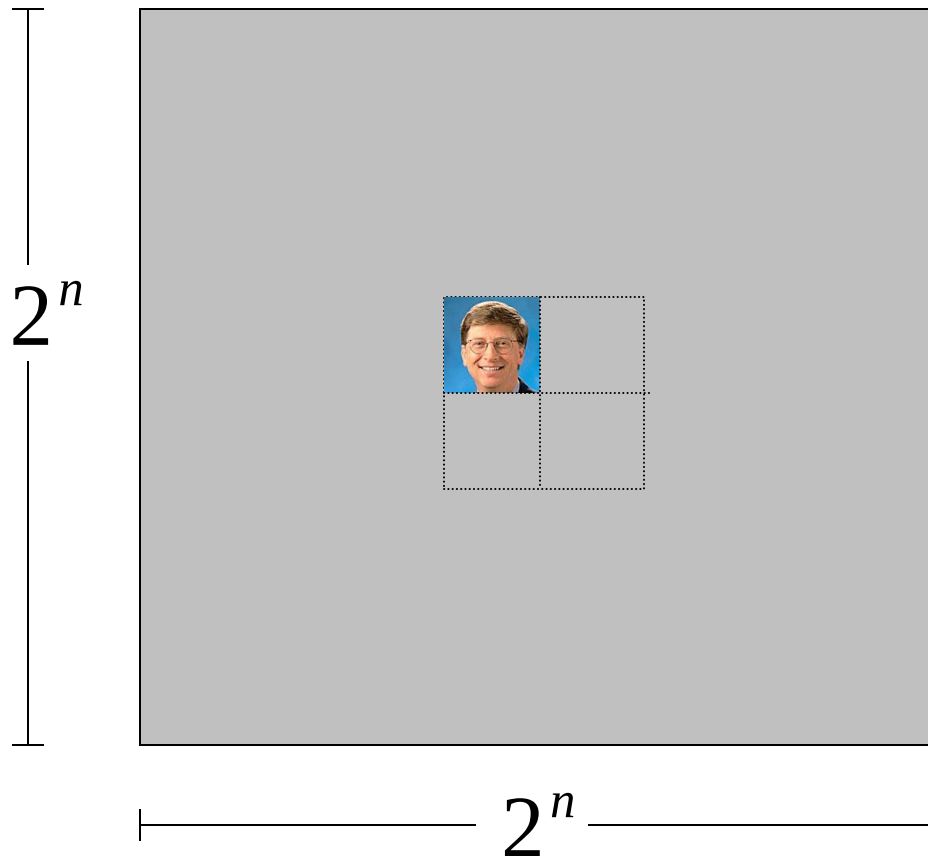
 $\sqrt{n+1} > \sqrt{n}$

This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
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Puzzle

Goal: tile the squares, except one in the middle for Bill.

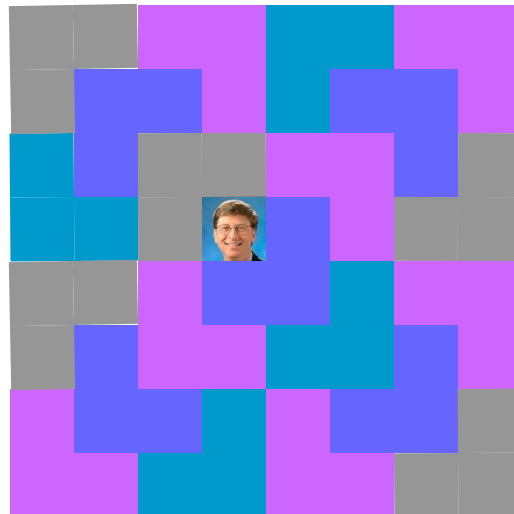


Puzzle

There are only trominos (L-shaped tiles) covering three squares:



For example, for 8 x 8 puzzle might tile for Bill this way:



Puzzle

Theorem: For any $2^n \times 2^n$ puzzle, there is a tiling with Bill in the middle.

Did you remember that we proved $2^{2n} - 1$ is divisible by 3?

Proof: (by induction on n)

$P(n) ::=$ can tile $2^n \times 2^n$ with Bill in middle.

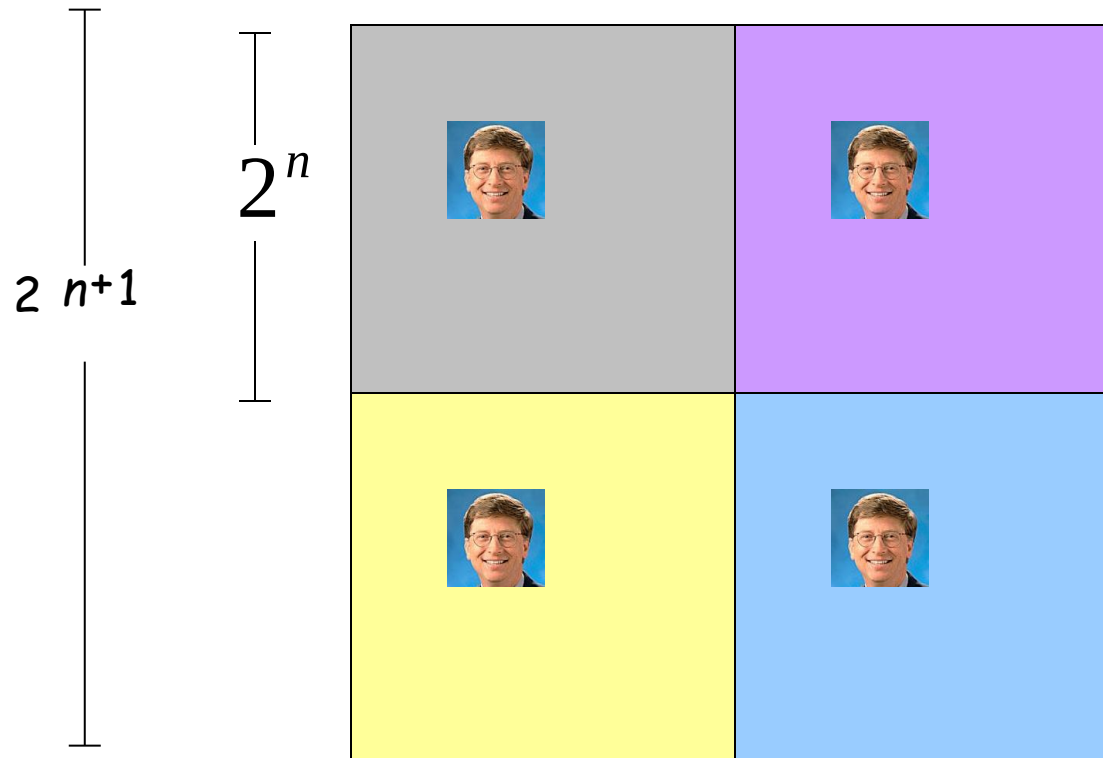
Base case: ($n=0$)



(no tiles needed)

Puzzle

Induction step: assume can tile $2^n \times 2^n$,
prove can handle $2^{n+1} \times 2^{n+1}$.



Puzzle

The new idea:

A stronger property

Prove that we can always find a tiling with Bill *anywhere*.

Theorem B: For any $2^n \times 2^n$ plaza, there is a tiling with Bill *anywhere*.

Clearly Theorem B implies Theorem.

Theorem: For any $2^n \times 2^n$ plaza, there is a tiling with Bill in the middle.

Puzzle

Theorem B: For any $2^n \times 2^n$ plaza, there is a tiling with Bill anywhere.

Proof: (by induction on n)

$P(n) ::=$ can tile $2^n \times 2^n$ with Bill anywhere.

Base case: ($n=0$)



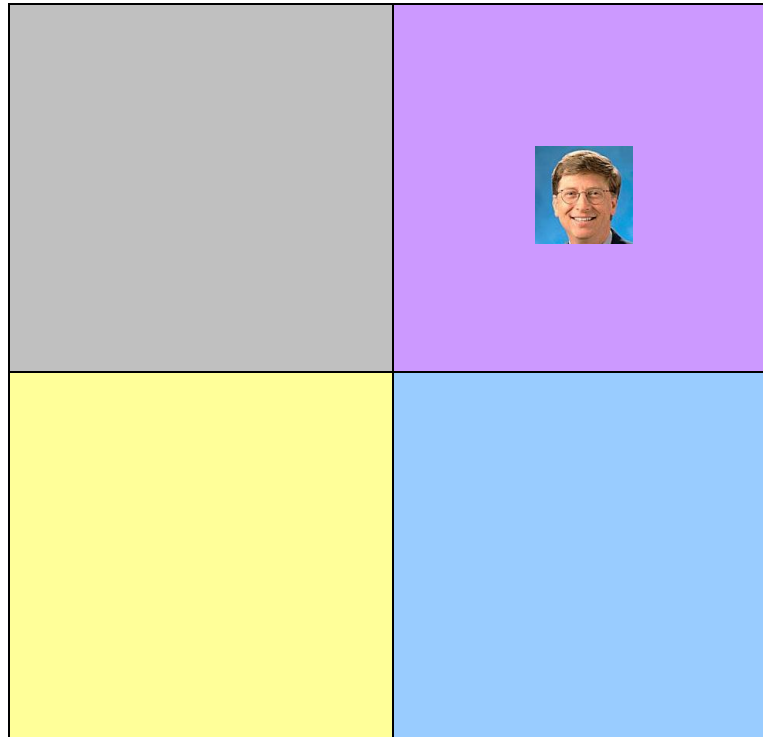
(no tiles needed)

Puzzle

Induction step:

Assume we can get Bill *anywhere* in $2^n \times 2^n$.

Prove we can get Bill anywhere in $2^{n+1} \times 2^{n+1}$.

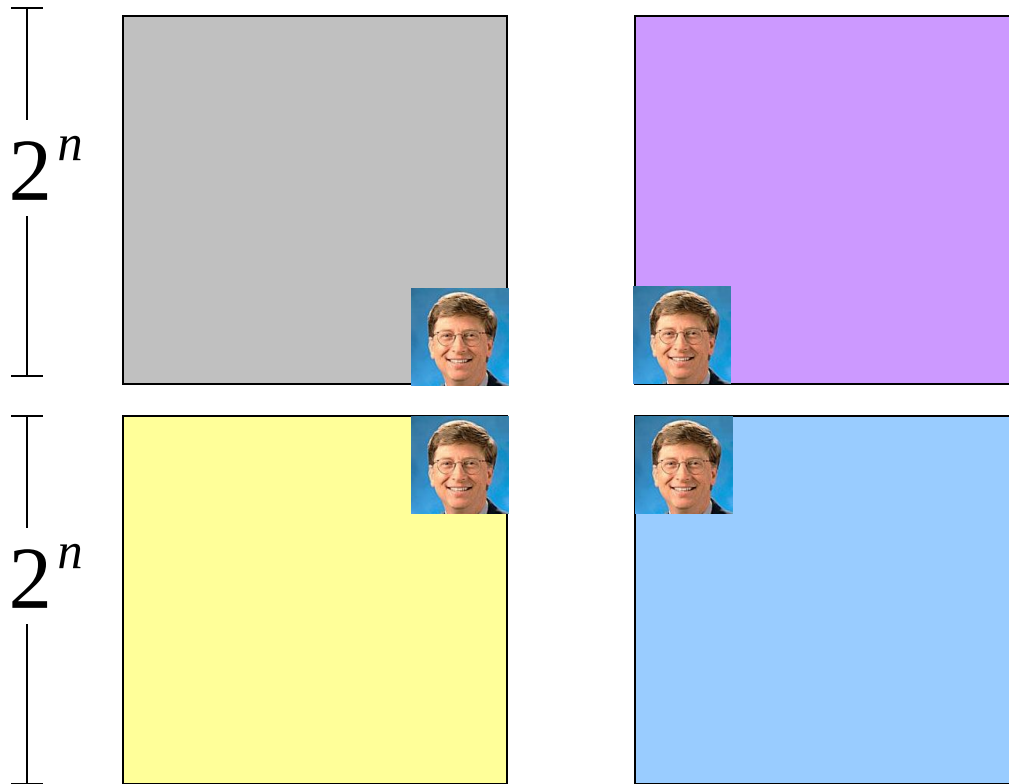


Puzzle

Induction step:

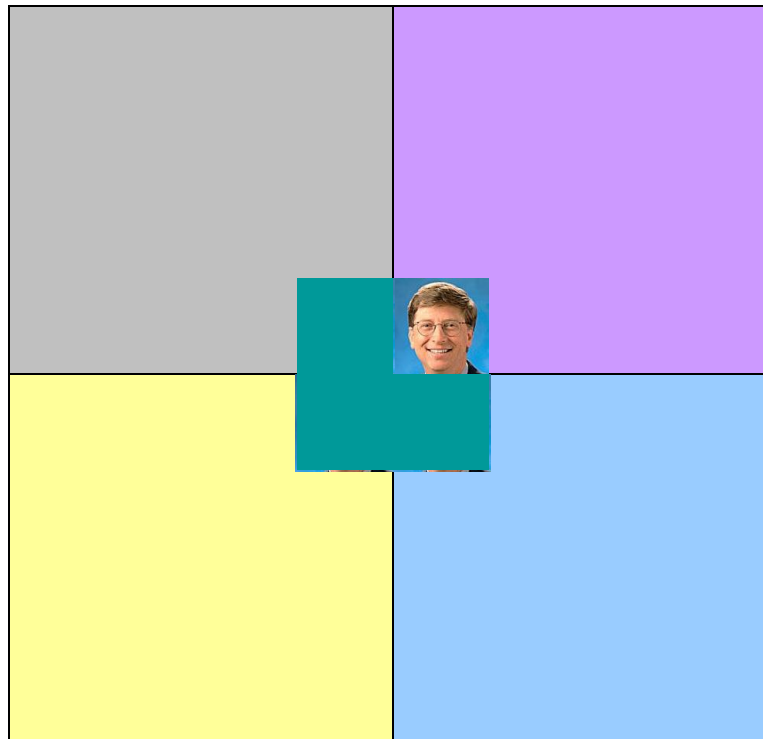
Assume we can get Bill *anywhere* in $2^n \times 2^n$.

Prove we can get Bill anywhere in $2^{n+1} \times 2^{n+1}$.



Puzzle

Method: Now group the squares together,
and fill the center with a tile.



Done!

This Lecture

- The idea of mathematical induction
- Basic induction proofs (e.g. equality, inequality, property, etc)
- An interesting example
- A paradox

Paradox

Theorem: All horses have the same color.

Proof: (by induction on n)

Induction hypothesis:

$P(n) ::=$ any set of n horses have the same color

Base case ($n=0$):

No horses so *obviously* true!

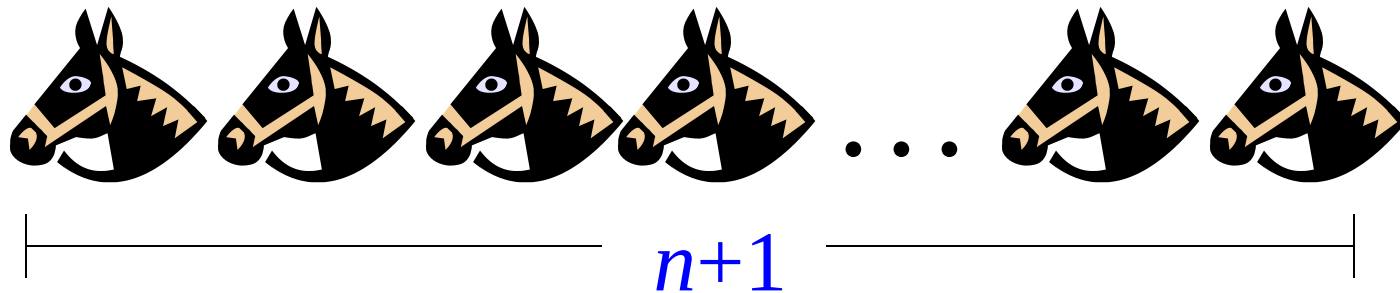


Paradox

(Inductive case)

Assume any n horses have the same color.

Prove that any $n+1$ horses have the same color.

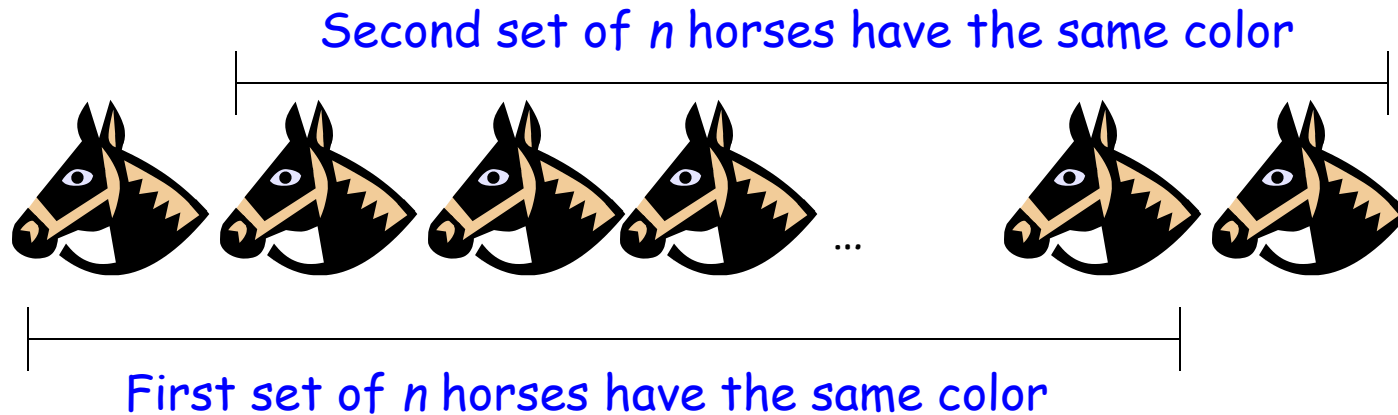


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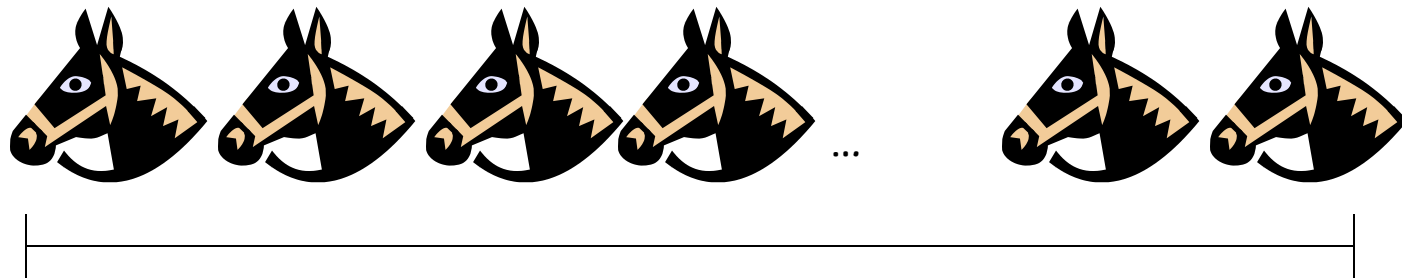


Paradox

(Inductive case)

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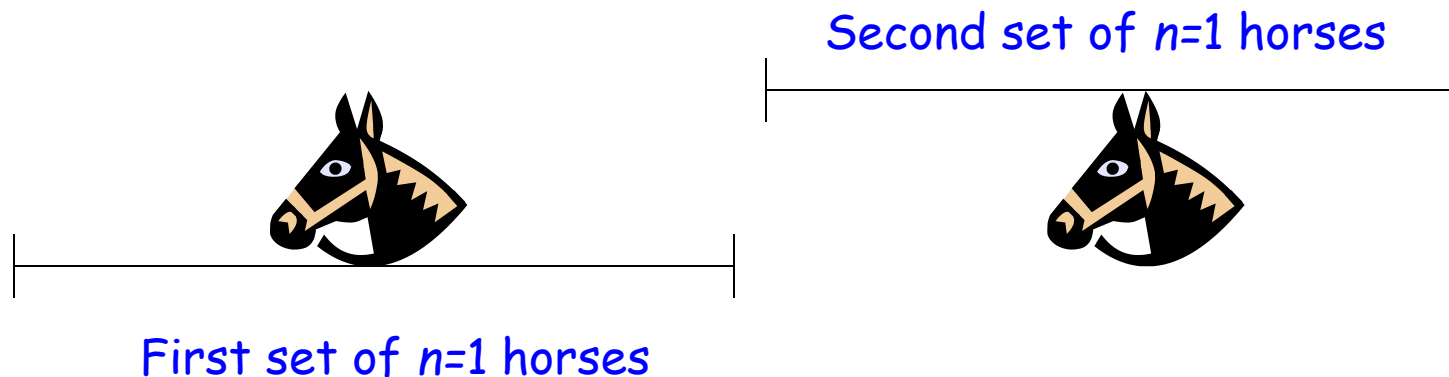


Therefore the set of $n+1$ have the same color!

Paradox

What is wrong? $n = 1$

Proof that $P(n) \rightarrow P(n+1)$
is *false* if $n = 1$, because the two
horse groups do not overlap.



(But proof works for all $n \neq 1$)

*The main problem is that you have assumed a overlap between any to sets;
This is NOT true*

Quick Summary

You should understand the principle of mathematical induction well,
and do basic induction proofs like

- proving equality
- proving inequality
- proving property

Mathematical induction has a wide range of applications in computer science.

In the next lecture we will see more applications and more techniques.